

SANDBERG, CA, USA

WMO: 723830

Lat: 34.744N

Lon: 118.724W

Elev: 4510

StdP: 12.45

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23187

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	26.6	28.3	-6.1	4.7	47.0	-1.2	6.1	46.5	40.1	36.5	35.6	37.7	15.4	340	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	93.1	61.0	90.4	60.2	87.9	59.2	65.7	84.5	63.6	83.4	61.9	82.1	12.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.3	92.4	72.0	56.2	79.6	71.9	53.7	72.6	71.0	32.9	84.4	31.1	83.7	29.8	81.9	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.3	28.6	25.7	DB	22.3	97.6	3.5	3.0	19.8	99.7	17.7	101.5	15.7	103.2	13.1	105.4	(4)
(5)			WB	19.1	69.3	3.8	5.8	16.4	73.5	14.2	76.9	12.1	80.2	9.4	84.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.9	44.9	44.2	48.5	51.2	58.6	68.2	75.6	75.8	70.6	60.5	51.1	43.9
	DBStd	14.48	7.93	8.76	9.17	9.82	9.62	9.21	5.96	5.98	8.42	9.33	9.33	7.82
	HDD50	1027	197	207	144	105	30	3	0	0	1	22	98	220
	HDD65	3796	624	583	516	423	247	73	4	5	47	197	424	653
	CDD50	3893	38	45	96	141	296	550	795	800	620	348	131	33
	CDD65	1187	0	1	4	9	48	170	334	340	216	58	7	0
	CDH74	9865	0	3	13	50	345	1491	3037	3023	1599	292	12	0
	CDH80	3743	0	0	0	3	86	553	1244	1246	557	54	0	0
Wind	WSAvg	13.4	14.1	14.2	14.0	14.5	14.4	13.9	11.8	11.5	11.5	12.9	13.4	14.5
Precipitation	PrecAvg	14.5	2.8	4.0	2.4	0.9	0.4	0.1	0.1	0.1	0.2	0.7	1.2	2.2
	PrecMax	29.8	12.2	14.4	8.3	3.0	3.8	1.1	1.2	0.2	1.0	4.4	3.2	8.0
	PrecMin	5.6	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0
	PrecStd	7.7	3.4	3.7	2.0	0.9	0.7	0.2	0.3	0.0	0.2	0.9	0.8	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.6	72.0	75.7	79.4	86.9	95.9	96.6	96.1	92.7	85.3	75.5	67.2
		MCWB	45.7	48.2	50.8	52.6	58.6	61.2	62.3	63.0	60.2	54.7	50.8	45.3
	2%	DB	61.6	64.9	69.9	74.4	82.2	90.5	93.0	92.3	89.2	80.8	71.7	61.5
		MCWB	43.8	45.8	49.1	51.1	56.6	59.3	61.4	60.8	58.9	54.1	48.4	43.6
	5%	DB	58.2	60.9	65.7	70.2	77.7	86.2	90.0	89.9	86.3	77.1	67.3	57.8
		MCWB	41.7	43.6	47.7	50.5	55.1	58.2	60.8	60.0	57.5	52.3	46.8	41.9
	10%	DB	55.5	57.0	61.9	65.8	73.4	82.3	87.0	87.2	82.9	73.4	63.7	54.6
		MCWB	40.6	41.8	45.7	48.7	53.6	57.8	59.8	59.0	56.7	51.0	45.3	40.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	50.7	53.2	58.6	64.5	69.6	69.2	68.4	66.1	60.9	53.9	51.9
		MCDB	55.9	64.1	68.4	68.0	71.6	80.8	86.4	87.4	80.8	74.9	68.7	55.6
	2%	WB	47.3	47.9	51.0	54.2	59.0	62.8	65.5	64.4	63.0	57.5	50.6	48.2
		MCDB	54.4	60.0	65.9	67.7	76.9	84.9	85.8	85.4	79.2	72.6	65.4	54.3
	5%	WB	45.3	45.7	49.3	51.8	56.8	60.6	63.8	62.3	60.9	55.0	48.9	45.8
		MCDB	52.2	56.4	62.1	66.3	74.6	81.7	83.9	84.7	78.8	71.3	62.6	52.4
	10%	WB	43.5	43.8	47.4	50.0	54.8	58.9	62.0	60.7	58.9	53.0	47.1	43.6
		MCDB	51.0	53.3	58.9	63.7	71.4	79.1	81.7	83.0	78.2	69.2	59.6	50.7
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.7	15.1	16.9	18.5	20.2	20.3	20.4	19.2	16.5	13.6	12.3
		MCDBR	14.3	16.7	18.4	20.5	21.7	22.4	21.6	21.1	20.3	18.5	15.9	15.7
		MCWBR	8.3	8.9	9.7	10.4	10.1	9.3	9.0	8.8	8.3	8.2	7.8	8.7
	5% WB	MCDBR	12.6	14.9	16.9	19.0	20.9	20.6	19.7	19.8	19.5	17.7	15.2	13.5
		MCWBR	8.5	8.9	9.7	10.9	10.7	10.7	10.2	10.0	9.6	9.3	8.7	9.5
Clear-Sky Solar Irradiance	taub		0.260	0.269	0.280	0.309	0.318	0.317	0.336	0.318	0.311	0.295	0.267	0.264
	taud		2.611	2.608	2.569	2.474	2.472	2.480	2.461	2.489	2.511	2.547	2.611	2.609
	Ebn at Noon		307	315	318	310	306	305	298	303	302	300	302	299
	Edh at Noon		23	26	30	35	35	35	35	34	31	28	23	22
All-Sky Solar Radiation	RadAvg		955	1238	1694	2102	2380	2599	2508	2355	1985	1485	1103	874
	RadStd		77	143	107	111	130	116	97	57	45	88	51	71

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.25	N/A	-3.04	+2.61	N/A	N/A	-216	-475	+616	+329
(47) Regional (49 neighbors)		+1.12	N/A	-2.97	+1.15	N/A	N/A	N/A	-218	+401	+167

Nomenclature: See separate page